(Please write your Enrolment No. immediately) Enrolment No. _____

MID-TERM EXAMINATION

B. TECH PROGRAMMES (UNDER THE AEGIS OF USCIT)

6th SEMESTER, MARCH,2025

Paper Code: FSD-318T **Subject: Advanced Java Programming**

Time: 1.5 Hrs. **Max. Marks :30**

Note: Attempt Q.No.1 which is compulsory and any two more questions from remaining.

Q.No.	Question	Max. Marks	CO(s)
1 (a)	Explain the key features of Java that make it a platform-independent language.	2	1
(b)	Describe the working of an applet in Java. How is it different from a standalone application?	2	1
(c)	How is a server implemented in Java? Discuss with an example.	2	1
(d)	How do stateful and stateless session beans differ?	2	3
(e)	Describe the role of HttpServletRequest and HttpServletResponse in servlet programming.	2	2
2 (a)	Write a Java program to demonstrate multithreading using the Runnable interface. Explain its execution flow.	5	1
(b)	Explain socket programming in Java with an example. How does a client communicate with a server using sockets?	5	1
3 (a)	Write a Java Servlet program to demonstrate session tracking using cookies. Explain how cookies help in maintaining session states.	5	2
(b)	Compare and contrast Servlets and JSP. Discuss their advantages and disadvantages in web development.	5	2
4 (a)	Explain the concept of exception handling in Java with an example. How does the try-catch-finally block work?	5	1
(b)	Write a Java program to create a simple Java Bean with properties and methods. Explain the importance of Java Beans in component-based development.	5	3

Enrollment No:.....

Supplementary Examination – March 2025

Programme: B.Tech[CSE/IT/CST/ITE/CS(AI)/CS(AIML)/CS(DS)] **Semester: 4rd Semester**

Paper Code: CIC-212 **Paper Name: Programming in Java**

Time: 1½Hrs. **Maximum Marks: 30**

Note:
- Question No. 1 is compulsory.
- Attempt any two questions from the remaining questions.
- Some questions have internal choice also.
- All questions carry equal marks.

	Question 1	Marks
1(a)	Can a Java interface have a constructor? Why or why not?	[2]
1(b)	Explain the purpose of class member modifiers public, private, protected in Java.	[2]
1(c)	Discuss the concept of method overriding in Java.	[2]
1(d)	Explain the purpose of nameless object in java.	[2]
1(e)	Explain the concept of Java applets and their typical use cases?	[2]
	Question 2	
2(a)	Explain the use of final keyword in java?	[2]
2(b)	Compare and contrast the wrapper classes for different primitive data types?	[4]
2(c)	Explain the methods provided by the StringBuffer class, such as append(), insert(), delete(), and reverse(). Provide examples demonstrating their usage.	[4]
	Question 3	
3(a)	Discuss the difference between JDK, JRE, and JVM.	[2]
3(b)	Explain the concept of "type casting" in Java. What are the differences between implicit and explicit casting?	[3]
3(b)	Provide examples to illustrate inheritance, polymorphism, encapsulation, and abstraction.	[5]
	Question 4	
4(a)	Discuss the security promises of the JVM. How does the JVM protect Java applications from various types of vulnerabilities?	[5]
4(b)	Describe how user-defined exceptions can be created in Java. Provide an example where a custom exception is used to handle specific application errors.	[5]

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(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES, 4TH Semester, March, 2025

Paper Code: CIC-212 **Subject: Programming in Java**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question (Attempt any 5 parts of Q1)	Max. Marks	CO(s)
1 (a)	What is wrapper class and name the wrapper class methods to convert string objects to primitive int.	2	CO 2
1 (b)	What is the sandbox model in java?	2	CO 3
1 (c)	What are the key characteristics of Java that make it platform-independent?	2	CO 1
1 (d)	What is the role of static keyword in java language?	2	CO 3
1 (e)	Difference between super and this keyword.	2	CO 2
1 (f)	What is garbage collection in Java, and how does it work?	2	CO 1
1 (g)	What is the difference between abstract class and interface?	2	CO 4
2 (a)	What is JVM and explain various components of JVM architecture? OR Explain the uses of the final keyword with suitable Java program.	5	CO 1
2 (b)	Explain the types of constructors in Java with suitable example. OR What is exception handling in Java and its advantages? Differentiate between throw and throws keyword in Java?	5	CO 2
3 (a)	Explain the life cycle of an Applet with suitable example. OR What are control flow statements in java, explain with suitable examples?	5	CO 3
3 (b)	What is an interface in java and how it is different from a class? OR Differentiate between method overloading & method overriding with an example.	5	CO 4

(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

B. Tech (CSE/IT/CS1/ITE) 4th Semester, April, 2024

Paper Code: __ CIC-212__ **Subject:** Programming in Java

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q.1 a. Explain the concept of garbage collection in java. [CO1]

b. Is JVM is an interpreter or emulator both? Discuss. [CO1]

c. Explain the concept of wrapper classes in Java and discuss their significance? [CO2]

d. Explain the use of keyword - Super. [CO2]

(2.5*4=10)

Q2. a. What is JVM? Explain JVM architecture in detail. **[CO1]** (7)

b. What is the sandbox model, and how does it help in providing security to resources? **[CO1]** (3)

Q3. a. Write a Java program demonstrating the implementation of an interface. Define an interface named Shape with methods calculateArea() and calculatePerimeter(). Implement this interface with classes Circle and Rectangle. Test the implementation by creating objects of these classes and invoking the interface methods to calculate their respective areas and perimeters. **[CO2]** (5)

b. What is Exception handling? Create a customized exception and also make use of all the 5 exception keywords. **[CO2]** (5)

Q4. a. Explain the security promises of the Java Virtual Machine (JVM) and how it achieves them? **[CO1]** (5)

b. Explain the concept of dynamic method dispatch and provide an application program to illustrate it. **[CO2]** (5)

(Please write your Exam Roll No.) Exam Roll No. .....

END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] JUNE 2024

Paper Code: CIC-212 **Subject: Programming in Java**

Time: 3 Hours **Maximum Marks: 75**

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit.

Q1 Answer the following questions: **(5x3=15)**

a) Discuss any five differences between Interface and Abstract Class.
b) Explain the concept of wrapper classes in Java and discuss their significance?
c) What is remote method invocation? Explain briefly.
d) How to create Inner & Outer Class? How it is helpful for achieving Security.
e) What is the thread's priority? Define the values for different priority in java.

UNIT-I

Q2 a) What is JVM? Explain JVM architecture in detail. **(8)**

b) What is Sandbox security model? How does it help in providing security to resources? Discuss in detail. **(7)**

Q3 a) Explain the security promises of the Java Virtual Machine (JVM) and how it achieves them. **(5)**

b) Write a short note on the following: **(10)**
(i)Various Instructions sets used in Java.
(ii)Garbage Collections
(iii)Class File Format

UNIT-II

Q4 a) Write a Java program demonstrating the implementation of an interface. Define an interface named Shape with methods calculate Area() and calculate Perimeter(). Implement this interface with classes Circle and Rectangle. Test the implementation by creating objects of these classes and invoking the interface methods to calculate their respective areas and perimeters. **(7)**

b) What is exception handling? Create customized exceptions and make use of all the 5 exception keywords used in exception handling. **(8)**

Q5. a) Explain the concept of Applet programming in Java and describe the life cycle of an applet. Provide a Java Applet program illustrating the key stages of an applet's life cycle. **(8)**

b) What is the difference between String Buffer and String Class? Show the use of two methods of String Buffer and String class with the help of example. **(7)**

P.T.O.

UNIT-III

Q6. a) What is thread? Explain the life cycle of a thread with the help of a diagram. What are the various ways to create the thread? Explain to anyone with the help of example. **(8)**

b) What is thread synchronization? When do we use it? Write a program to illustrate the concept of Thread Synchronization method **(7)**

Q7. a) Why is event used in Java? What is the difference between adapter class and listener interface? Explain with the help of an example. **(8)**

b) Discuss various types of layouts used in AWT. Write a program to show the use of Border Layout and Grid Layout with various components. **(7)**

UNIT-IV

Q8. a) Explain different types of JDBC drivers. Write the steps to connect a Java application to a database? Explain with the help of example. **(10)**

b) Write a program to sort the content of a given text file. **(5)**

Q9. a) Write a program to implement the Client -Server Application. **(7)**

b) Write the short note on: **(8)**
(i)What is JNI and how it works?
(ii)What is object serialization vs deserialization?
(iii)What are Java collections? Give some methods of Collection interface.

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